

CS 2300 – Homework 6

Submit a digital copy (of PDF) to Canvas

Total possible points: 100 points

Homework Guidelines

No Handwritten Answers

This homework will require some written responses and some diagrams. For the written responses, use a standard word processor like Microsoft Word, LibreOffice Writer, Google Docs, or LaTeX. Drawings can be made using Microsoft PowerPoint or Google Drawings¹, among others (e.g. draw.io, lucidchart). You can explore other tools as well, just search around and verify the tool you are considering is compliant with the design aspects that are presented in class.

Please create your answers in a digital format, export to a PDF, and upload that PDF to Canvas. No need to turn a printed copy. For those of you considering LaTeX, ShareLaTeX and Overleaf are good online editors.

One Problem per Page

In the past, many students turned in a single piece of paper with all answered questions smooshed onto it. This did not leave my grader or I with adequate room to provide suggestions on how a design could be improved nor explanations on errors. The process of design work is to work in iterative stages: the first stage will be rather messy (and can be hand-written), the second stage will have things cleaned up, and so on. The final stage is what you should turn in, and it should be free of scratched-out work and high-density drawings. Only answer one problem on *at least* one page, and refrain from super-compact answers that leaves more than 1/2 of the page empty. Ideally, leave somewhere between 1/4 to 1/2 of a page empty so that we can make annotations.

¹ <https://docs.google.com/drawings/>

Problem 1 [15 pts]

Consider the following relation $R\{A, B, C\}$:

A	B	C
10	B1	C1
10	B2	C2
11	B4	C1
12	B3	C4
13	B1	C1
14	B3	C4

a. Given the previous extension (state), which of the following dependencies may hold in the above relation? If the dependency cannot hold, explain why by specifying the tuples that cause the violation.

- i. $A \rightarrow B$ ii. $B \rightarrow C$ iii. $C \rightarrow B$ iv. $B \rightarrow A$ v. $C \rightarrow A$

b. Does the above relation have a potential candidate key? If it does, what is it? If it does not, why not?

Problem 2 [15 pts]

Consider the following two sets of functional dependencies:

$$F = \{A \rightarrow C, AC \rightarrow D, E \rightarrow AD, E \rightarrow H\} \text{ and } G = \{A \rightarrow CD, E \rightarrow AH\}.$$

Are they equivalent? Why or Why not?

Problem 3 [20 pts]

Consider the following relation R and functional dependencies F .

$$R = \{A, B, C, D, E, G, H, J\}$$

$$F = \{A \rightarrow BCD, E \rightarrow GH, C \rightarrow D, AE \rightarrow J\}$$

Compute the minimum cover of F . List all the three steps.

Problem 4 [25 pts]

Consider the following relation:

CAR_SALE(Car#, Date_sold, Salesperson#, Commission%, Discount_amt)

Assume that a car may be sold by multiple salespeople, and hence {Car#, Salesperson#} is the primary key. Additional dependencies are:

Date_sold $\rightarrow$ Discount_amt
Salesperson# $\rightarrow$ Commission%

Based on the given primary key, is this relation in 1NF, 2NF, or 3NF? Why or why not? How would you successively normalize it completely?

Problem 5 [25 pts]

Consider the relation REFRIG(Model#, Year, Price, Manuf_plant, Color), which is abbreviated as REFRIG(M, Y, P, MP, C), and the following set F of functional dependencies:

$F = \{M \rightarrow MP, \{M, Y\} \rightarrow P, MP \rightarrow C\}$

- a. Evaluate each of the following as a candidate key for REFRIG, giving reasons why it can or cannot be a key: {M}, {M, Y}, {M, C}.
- b. Based on the above key determination, state whether the relation REFRIG is in 3NF and in BCNF, and provide proper reasons.
- c. Consider the decomposition of REFRIG into $D = \{R_1(M, Y, P), R_2(M, MP, C)\}$. Does this decomposition preserve all of the functional dependencies? Is this decomposition lossless? Show why. (Use the *Nonaddictive (lossless) join test*)